



EAST WEST UNIVERSITY

Physics Lab
Department of MPS

Course Name : Engineering Physics -1

Course Code : PHY 109 LAB

Section No : 08

Group No : 05

Experiment No : ~~To det~~ 12

Name of the Experiment : To determine modulus of rigidity of
a wire by the method of oscillations
(Dynamic method)

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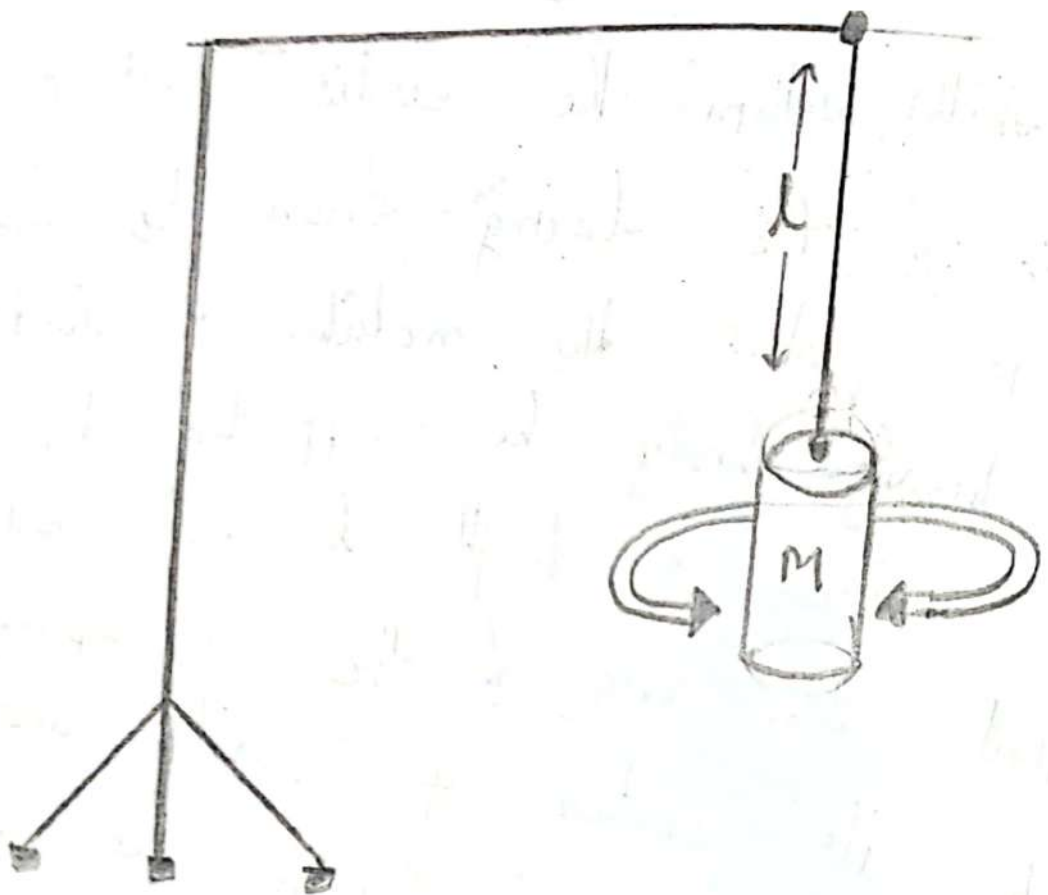
PHY-109 LAB

Experiment No : 12

Name of the Experiment : To determine the modulus of rigidity of a wire by the method of oscillations (Dynamic method).

Theory : Within the elastic limit of a body, the ratio of the shearing stress to shearing strain is called the modulus of rigidity. If a heavy body be supported by a vertical wire of length l and radius r , so that the axis of the wire passes through its center of gravity and if the body be turned through an angle and released, it will execute torsional oscillations about a vertical axis. If at any instant the angle of twist be θ , the moment of the torsional couple exerted by the wire will be,

$$C = \frac{\pi \eta r^4 \theta}{2l}$$



When $\theta = 1$ radian

$$\text{Then, } C = \frac{\eta \pi r^4}{2l} \dots \dots (1)$$

Where, η is the modulus of rigidity of the material of the wire.

Therefore, the motion is simple harmonic and of the fixed period.

$$T = 2\pi \sqrt{\frac{I}{C}}$$

$$\text{or, } T^2 = \frac{4\pi^2 I}{C}$$

$$\text{or, } C = \frac{4\pi^2 I}{T^2} \dots \dots (2)$$

Where, I is the moment of inertia of the body.

$$\text{And } I = \frac{1}{2} M a^2$$

Where, M = Mass of the cylinder, a = radius of the cylinder.

From equation (1) and (2), we get

$$\frac{\eta \pi r^4}{2l} = \frac{4\pi^2 I}{T^2}$$

$$\text{or, } \eta = \frac{4\pi^2 I}{T^2} \times \frac{2l}{\pi r^4}$$

$$\text{or, } \eta = \frac{8\pi I l}{T^2 r^4} \text{ dynes/cm}^2 \dots (3)$$

putting the value of I , l , n and T ,
we can calculate n .

Apparatus :

1. A uniform wire
2. A disc on cylindrical bar
3. A suitable clamp
4. A stop - watch
5. A screw gauge
6. A slide calipers
7. A meter scale
8. A balance, etc.

Data Collection:

(A) Length of the wire, $l = 51.5$ cm

Calculation of the least count:

$$\text{Pitch (P)} = \frac{\text{The distance moved by plane of the disc along vertical scale}}{\text{Number of full rotations given to the circular scale}}$$

=

=

Number of total divisions in the circular scale, $n = 100$

$$\begin{aligned}\text{Least count (L.C.)} &= \frac{\text{pitch}}{\text{Number of divisions of circular scale}} \\ &= \frac{1}{100} \\ &= 0.001 \text{ cm}\end{aligned}$$

(B) Reading for the Radius of the Wire (r)

No. of obs.	Linear Scale Reading (x) cm	Circular Scale Divisions (CSD)	Least Count (LC) cm	Diameter $d = x + (\text{CSD} \times \text{LC})$ cm	Mean Diameter (d) cm	Mean Radius $r = d/2$ cm
01	0	53	0.001	0.055	0.055	0.0275
02						
03						

(C) Reading for the Radius of the Cylinder (a)

No. of obs.	Main Scale Reading (x) cm	Vernier Scale Division (VSD)	Vernier Constant (VC) cm	Diameter $D = x + (VSD \times VC)$ cm	Mean Diameter D cm	Mean Radius $a = D/2$ cm
01	5	1	0.005	5.005	5.005	2.5025
02						
03						

(D) Reading for the Time Period (T)

No. of obs.	Time for 20 Oscillations (t) sec	Time Period $T = (t/20)$ sec	Mean Time Period T sec
01	31.42	3.142	3.142
02			
03			
04			
05			

Calculation:

Mass of the cylinder, $M = 104.5$ gm

Moment of inertia of the cylinder, $I = \frac{1}{2} M a^2$

$$= 3272.16 \text{ gm-cm}^2$$

$$= \text{gm-cm}^2$$

From data collection,

$$l = 51.5 \text{ cm}, r = 0.0275 \text{ cm}, T = 1.536 \text{ sec}, I = 3272.16 \text{ gm-cm}^2$$

So, we get from equation (3)

$$\eta = \frac{8\pi l}{T^2 r^4}$$
$$= \frac{7.5 \times 10^{11}}{\text{dynes/sq.cm}}$$
$$= \text{dynes/sq.cm}$$

Result:

The Modulus of rigidity of a wire is 7.5×10^{11} dynes/sq.cm

Precautions & Discussions:

- (i) There should be no knot in the wire.
- (ii) The amplitude of oscillation should be small so that wire is not twisted beyond the elastic limit.
- (iii) To avoid the backlash error, the circular scale reading of screw gauge should be taken very carefully.
- (iv) The radius of the suspension wire occurs in the 4th power and hence it should be measured very carefully.

Sample Oral Questions and Answers

1. What is modulus of rigidity?

Ans: The modulus of rigidity of an object is a measure of the resistance to deformation of an object when an external force is applied. It is expressed as the ratio of shear stress to shear strain.

2. What are the factors affecting modulus of rigidity?

Ans: The factors that affect modulus of rigidity are the material properties and the applicable force over the material.

3. What is an oscillation?

Ans: The repeated back and forth movement between two positions or states of an object is known as oscillation. The motion of the body is said to be oscillatory or vibratory motion if it moves back and forth about a fixed position or point after a regular interval of time.

4. What is moment of inertia?

Ans: Moment of Inertia is a quantity which expresses the body's tendency to resist external angular acceleration. It can be calculated as the sum of the products of the mass of each particle of the body with the square of its distance from the axis along which it is to be rotated.

Result:

The Modulus of rigidity of a wire
is 7.5×10^{11} dynes/sq.cm

Precautions and Discussions:

1. Ensure the wire is free from knots, kinks, or any imperfections that could be affect its rigidity.
2. Maintain small oscillation amplitudes to avoid exceeding the elastic limit of the wire.
3. Take accurate measurements of the wire's length and diameter.
4. Use a precise timer to accurately measure the time period of oscillations.
5. Conduct the experiments in a stable environment, minimizing air currents and temperature fluctuations that could influence the wire's properties.